

Worksheet 6 – Hierarchical Clustering (Single linkage method)

Suppose the daily amount spent on meals (X_1) and tuition (X_2) of five students are shown below (data are made up, and used here just as an example):

Student	X_1	X_2
A	2	4
B	8	2
C	9	3
D	1	5
E	8.5	1

(a) Plot the observations in a scatterplot (scatter diagram). How many groups would you think there are, and what are the members?

(b) Apply the single linkage method, using Euclidean distance as the distance measure.

(c) Draw a dendrogram, and use it to deduce the number of groups (clusters) and the membership.

(d) Write R code to validate that your dendrogram and the heights are correct.